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(54) Title of the invention : METAMATERIAL UNIT-CELLS LOADED GROUND-STRUCTURE AND RADIATING-PATCH MICROSTRIP-ANTENNA FOR INTELLIGENT-TRANSPORTATION-SYSTEM APPLICATIONS IN S-, C-, X-, KU-BAND

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(57) Abstract :

ABSTRACT TITLE: METAMATERIAL UNIT-CELLS LOADED GROUND-STRUCTURE AND RADIATING-PATCH MICROSTRIP-ANTENNA FOR INTELLIGENT-TRANSPORTATION-SYSTEM APPLICATIONS IN S-, C-, X-, KU-BAND The present invention relates to a metamaterial unit cell loaded radiating patch and defected ground structure microstrip patch antenna (MUCRPDGS) (100) for Intelligent Transportation System applications. The antenna (100) comprises a three-layer structure including a top radiating patch (103) with concentric split ring resonators, a middle FR4 substrate (102), and a defected ground structure (101) with square split ring resonators. The antenna has compact dimensions of $3.2 \times 2.8 \times 0.16 \text{ cm}^3$. The top layer includes unit cells with outer dimensions of $4 \text{ mm} \times 4 \text{ mm}$ and inner rings of radii 4.94 mm, 4.04 mm, 1.0 mm, and 0.75 mm, while the ground plane includes $1.5 \text{ mm} \times 1.5 \text{ mm}$ unit cells. The antenna exhibits triple broadband characteristics with bandwidths of 0.46 GHz (3.20–3.66 GHz), 4.52 GHz (6.83–11.35 GHz), and 4.03 GHz (13.52–17.55 GHz), along with enhanced gain and compact structure. [Fig.1]

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